SYSTEM ARCHITECTURE OF STREAMSHIELD

Live broadcasting, resilient delivery and independently verified recovery

CREATOR
Camera / screen → browser Studio, or OBS

AUDIENCE
Watch / discovery · HLS playback · live chat

ADMINISTRATOR
Control · health · incidents · logs · chaos

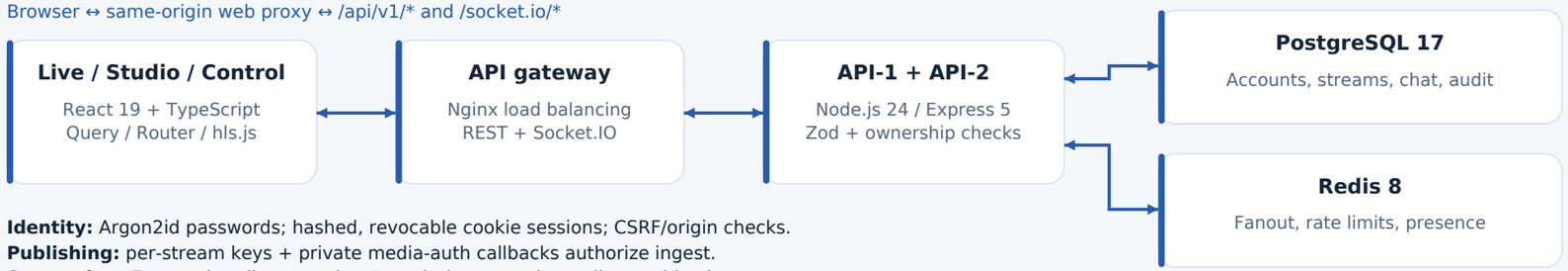
01 MEDIA PLANE · capture → ingest → delivery

Creator ingest: WHIP signaling uses /ingest/* via web; WebRTC ICE media goes directly to the relay. OBS uses RTMP.

Replication: origins pull private authenticated RTSP. Primary/standby HLS paths remain distinct. MediaMTX remuxes; no ABR transcoder.

02 APPLICATION + DATA PLANE · metadata and chat

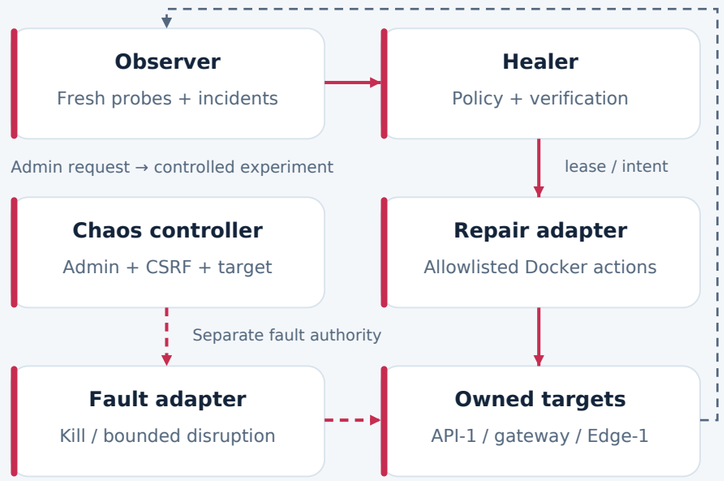
Browser ↔ same-origin web proxy ↔ /api/v1/* and /socket.io/*



Identity: Argon2id passwords; hashed, revocable cookie sessions; CSRF/origin checks.
Publishing: per-stream keys + private media-auth callbacks authorize ingest.
Separation: Express handles metadata/permissions, not the audience video bytes.

03 RELIABILITY CONTROL

Independent processes; evidence drives every state



SQLite: durable intents, leases and budgets.
PostgreSQL: incident, recovery and experiment history.
Control UI: authoritative REST + Socket.IO invalidations.
Proof: healthy probes + application/HLS behavior; not a PID.
Network case: Toxiproxy → simulated read-only dependency.

04 FAILURE → DETECTION → RESTORATION → VISIBLE EFFECT

RTO = injection to independent service verification

FAILURE	DETECTED BY	RESTORATION + VERIFICATION	LIVE-SYSTEM EFFECT	LOCAL RTO
API server crash	Readiness unreachable; confirmed probe failures.	Healer starts API-1; readiness, catalog and gateway routing prove return.	API-2 serves; video can keep playing.	21.500 s
Video gateway crash	Gateway probe and actual HLS delivery fail.	Healer starts the gateway; valid, advancing HLS segments verify it.	Buffer may drain, then playback resumes.	15.324 s
Edge node failure	Edge-1 probe fails; Edge-2 delivery observed.	Healer starts Edge-1; live media verifies repair; peer stays unchanged.	Edge-2 failover can preserve video.	14.232 s
Network disruption	Toxiproxy delays safe recommendation reads; circuit opens.	Owned 20 s delay expires; successful HALF_OPEN reads verify reconnection.	Recommendations impaired; video separate.	27.247 s
Media authority loss	Both HLS routes deny requests; buffered video drains.	Operator restores simulated authority; advancing media must verify return.	Video stops. Human restoration required.	Unmeasured **MANUAL**

Measured run a484d151: Linux + generated live source; timings are not Windows guarantees or viewer stall durations. Manual case: separate acceptance evidence.

DEPLOYMENT + TRUST BOUNDARY

Docker Compose on one host: **13-container compact demo**. API-2 and control roles share one container; restricted API-2 cannot use its Docker socket. Only trusted adapters act.
5173: persistent base app · **5174:** isolated ss-light-* streaming/chaos fixture. Faults stay in the owned demo project; the base app has no Healer by default.

GUARDS + OPTIONAL FULL TOPOLOGY + LIMITS

Default: **2 repair operations/service/10 min + 30 s cooldown**. No restart of the last serving API; no automatic PostgreSQL repair. Healer is policy code, not an LLM.
Optional full profile: Prometheus, Grafana, exporters and spare API-3. Single host/singletons remain limits; public deployment needs TLS, NAT/TURN, secrets and off-host recovery.